

Item screening in graphical Rasch models. A tutorial

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Abstract

Rasch models are widely used to analyze questionnaire and test data. Inference based on the standard Rasch model is only valid when the data meet assumptions such as local independence and absence of differential item functioning. Graphical Rasch models provide a framework where local dependence and differential item functioning can be represented explicitly. In this context, "item screening" denotes a coherent workflow for testing for consistency, local dependence, differential item functioning, score-covariate associations, and graph-based model checking. This is done while taking spurious evidence into account. We present a non-technical introduction to the screening procedure and describe an implementation in R.

1 Introduction

Questionnaires and tests are widely used to measure health, abilities, and experiences. This makes it essential to ensure that they produce reliable, meaningful, and fair results. This can be done by evaluating whether observed data meet the requirements of the Rasch (1960) model. A model that satisfies all formal psychometric criteria of the Rasch model can be considered the gold standard of measurement ensuring validity and objectivity.

Among these are the assumptions that there is no local dependence (Lazarsfeld and Henry, 1968, LD) and that there is no differential item functioning (Holland and Wainer, 2012, DIF).

Briefly, local independence means that items are only correlated through the latent trait that the instrument measures Lord and Novick (1968) and DIF that respondents from different groups who have the same level of the latent variable have different response distributions on an item.

Graphical Rasch models Kreiner and Christensen (2002, 2004, 2007) provide a framework for including item dependencies and differential item functioning. This class of models does not satisfy all measurement requirements, but offers a realistic and practically useful representation of empirical data Kreiner (2007).

Identifying explicit violations of the Rasch model requirements based on empirical data is essential. Each item pair can be tested for LD using several available methods (Christensen et al., 2017, pp. 179-80). Similarly, several statistical tests exist that can evaluate of DIF for each combination of an item and a covariate Zumbo (1999).

In the context of LD, Kreiner and Christensen (2012) discuss two central testing approaches: conditional likelihood ratio tests Kelderman (1984) and a simple non-parametric test that does not require specialized Rasch software (Kreiner and Christensen, 2004, pp. 1250-1). These also apply to tests of DIF.

The latter of these approaches is the basis of a stepwise screening procedure for simultaneously evaluating LD and DIF Kreiner and Christensen (2011). This procedure involves checks of consistency, tests of LD, tests of DIF, elimination of spurious evidence, and graph-based checks of the resulting model structure.

1.1 Models for measurement

The Rasch model describes the relationship between a latent trait, denoted θ , and observed item responses $\mathbf{Y} = (Y_i)_{i=1,\dots,k}$, where $Y_i \in \{0, 1, \dots, M_i\}$. For ease of notation, we assume that $M_i = M$ for all items i , i.e. that all items have the same number of response options. The latent trait θ is the quantity or trait that the instrument intends to measure. The Rasch model imposes the following requirements: (i) θ is uni-dimensional; (ii) local independence: $Y_i \perp Y_j | \theta$ for all item pairs (X_i, X_j) ; (iii) Absence of DIF: $Y_i \perp X_l | \theta$ for all pairs (Y_i, X_l) where Y_i is an item and X_l is a covariate; (iv) the parametric form of the response probabilities is

$$P(Y_i = y | \theta) = \frac{\exp(y\theta + \alpha_{iy})}{\sum_{z=0}^M \exp(z\theta + \alpha_{iz})}$$

The last requirement ensures that the item response function (IRF) $\theta \mapsto E(X_j = x | \theta)$ is monotone and increasing.

1.1.1 Notation

For each participant total scores are computed as $S = \sum_{i=1}^k Y_i$ and rest-scores as $R_i = S - Y_i$. Furthermore, for any subset of items $B \subset \{Y_1, \dots, Y_k\}$ we define sub-scores $S_B = \sum_{Y_i \in B} Y_i$.

1.1.2 The graphical Rasch Model

Combining the requirements (ii)-(iv) we can write the Rasch model in vector form

$$P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x}, \theta) = \exp \left(\alpha_0 + \theta S + \sum_{i=1}^k \alpha_{iy_i} \right) \quad (1)$$

where

$$\alpha_0 = -\log \left(\prod_{i=1}^k \sum_{z=0}^M \exp(z\theta + \alpha_{iz}) \right)$$

is a normalizing constant that does not depend on the observed data $(\mathbf{y}, \mathbf{x})$. It is a consequence of (1) that S is sufficient for θ . This is a feature that makes the Rasch model stand out from other models for measurement. Explicitly, this means that under the Rasch model, the total score S , contains all information in the response patterns, that is relevant for estimating θ . However, it is important to specify that S and θ are not identical objects, but S is an observed statistic, which holds the necessary information about the unobserved parameter θ , under the Rasch model.

1.1.3 Adding LD and DiF

The general form represented in (1) is a log-linear model containing only main effects, and this makes it easy to include (i) interaction terms between item pairs to model LD and (ii) to include interaction terms between item and exogenous variable pairs to model DIF. This leads to the following form for the GLLRM for polytomous items

$$P(\mathbf{Y} = \mathbf{y} | \Theta = \theta, \mathbf{X} = \mathbf{x}) = \exp \left(\alpha_0 + \theta S + \sum_{i=1}^k \alpha_i + \sum_{i=1}^{n_L} \lambda_i(u_i, v_i) \right) \quad (2)$$

$$+ \sum_{i=1}^{n_D} \delta_i(a_i, z_i) + \sum_{i=1}^{n_G} \mu_i(g_i), \quad (3)$$

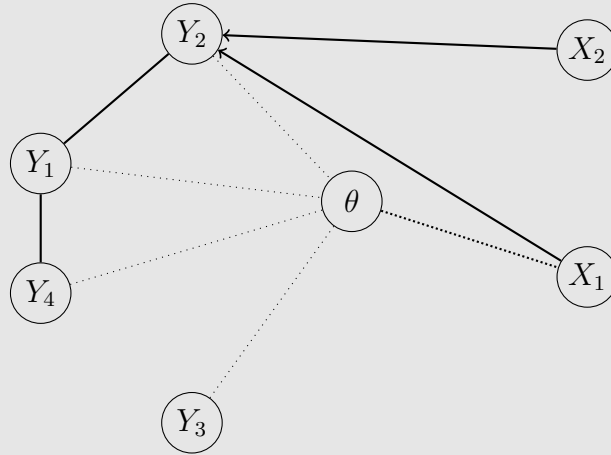
where n_D denotes the total number of DIF pairs $D = (D_1, \dots, D_{n_D})$ with $D_i = (A_i, Z_i)$, where $A_i \in \{Y_1, \dots, Y_k\}$, and $Z_i \in \{X_1, \dots, X_m\}$ is a source of DIF for A_i . n_L denotes the total set of LD pairs with $L = (L_1, \dots, L_{n_L})$ with $L_i = (U_i, V_i) \subset$

$\{Y_1, \dots, Y_k\}$. Lastly, n_G is the total set of interactions with more complex DIF or LD structure, $G = (G_1, \dots, G_{n_G})$ with $G_i \subset \{Y_1, \dots, Y_k, X_1, \dots, X_m\}$ Kreiner and Christensen (2011).

1.1.4 Representation as graph

The model can be represented as a graph where items and covariates are represented as nodes. Edges between item nodes represent LD, while edges between item and exogenous variable nodes represent DIF.

Example 1.1. Assume we have a dataset with four items Y_1, Y_2, Y_3, Y_4 and two exogenous covariates X_1, X_2 . Suppose within the dataset there exists DIF between the following pairs $(Y_2, X_1), (Y_3, X_1), (Y_2, X_2)$, LD between $(X_1, X_2), (X_3, X_4), (X_1, X_4)$, and finally that X_1 is related to θ . This can be represented in the graph



2 Item Screening

The posed item screening we are implementing is to be found, described in Table 5 in Kreiner and Christensen (2011), but we will include it here for completeness.

Step 1	Analysis of consistency
Step 2	Analysis of DIF and LD
Step 3	(a) Elimination of spurious evidence of LD (b) Elimination of spurious evidence for items with more than one DIF source by tests of $Y_i \perp X_j \mid S, \text{Source}(Y_i)$ (c) Elimination of spurious evidence for sources with an apparent DIF effect on more than one item by tests of $Y_i \perp X_j \mid S, \text{DIF}(X_j)$
Step 4	Analysis of the association between the score and exogenous covariates
Step 5	Definition of a GLLRM and analysis of the global Markov properties of G_{Rasch}
Step 6	Tests of global Markov properties of the GLLRM

Table 1: The strategy for item screening presented in paper by Kreiner and Christensen (2011).

2.1 Under the Hood of the Item Screening Procedure and How It Is Implemented

A lot of information is embedded within Table 1, while we refer the reader to Kreiner and Christensen (2011) for a full derivation of the procedure, we will now unfold the steps, argue how we understand each one, and how the methods we use are implemented in the package.

2.1.1 Step 0: A Note on Pre-Processing

What do we do with missing values? One possibility is simply to throw away all observations that contain one or more missing values. When we encounter missing values in item responses we cannot compute total scores and therefore no rest-scores, which means we won't be able to test for DIF and LD, and the list of problems this entails goes on. However, it seems overkill to throw an entire observation away because a participant forgot to note their gender when filling out the questionnaire. The main goal of the pre-processing we force upon the data, after any user has processed the data themselves, is to keep as many observations when performing the different tests as possible. In all the steps we are working with item scores, so we need these to be complete; therefore, this is fairly strict. However, the distribution

of missing observations in exogenous variables might lead to an even thinner dataset if we also throw away these data points. Hence, we will only do so per exogenous variable. Meaning when we screen for DIF, we do so on a subset of the dataset that considers complete cases with respect to all items and the one exogenous variable only. This way we ensure as many observations as possible are used in each step, without introducing any new information to the data.

2.1.2 Step 1

In this initial step of the item screening, we are only concerned with associations. Association between variables can be interpreted many ways, but the objective is simply to determine the relationship between variables. We choose to work with the partial gamma coefficient as per Kreiner and Christensen (2011), but other measures are also possible. We use this association measure to check whether the data fulfill the three necessary assumptions M1, M2, and M3.

M1 demands the chosen measure of association is strictly non-negative between all pairs of items in the data set.

M2 demands that any item Y_a is positively related to its corresponding rest-score R_a , and all sub-scores S_B , where $Y_a \notin B$.

M3 demands that if an exogenous variable X is positively related to the latent, person-specific, variable θ , then X will also be positively related to the total score S , all sub-scores, S_A , and all item responses Y_i .

Step 1 in practice. In all of the consistency checks necessary in step 1, we are at no point controlling for a third variable; hence, the partial gamma coefficients are simply Goodman and Kruskal's gamma. M1 and M2 does not leave much haggling to be done, so in these two checks we simply compute the partial gamma coefficients, check they are all non-negative and note a successful pass if so. M3 states that, if an exogenous variable X is positively related to θ , then it will also be positively related to S , all S_A , and all Y_i . We interpret this as a demand which also must hold with reverse sign. Meaning that the direction of association between an exogenous variable X and all items Y_i must be the same as the direction of association between the same exogenous variable X and the total score S . Hence, we investigate this assumption by computing the partial gamma coefficient between X_j and S , and check that the direction of association between X_j and Y_i for $j = 1, \dots, m$ is the same, for all $i = 1, \dots, k$. If so, we note a successful pass of M3. Naturally, if the data passes M1, M2, and M3 we are happy and can safely move onto the next step. If the data fails any of these checks, we should re-evaluate our data and consider if we want to continue with the item screening.

2.1.3 Step 2

In this step, we begin the analysis of DIF and LD. We use the partial gamma coefficient to flag all item pairs exhibiting LD and all item and exogenous variable pairs that experience DIF. We test for these two things via hypotheses of the form

$$H_0^{DIF} : Y_i \perp X_j \mid S \quad \text{and} \quad H_0^{LD} : Y_i \perp Y_{i'} \mid R_i,$$

Kreiner and Christensen (2011). Rejecting the null if the corresponding asymptotic p -value of the partial gamma coefficient, after controlling for multiple testing using the Benjamin-Hochberg procedure, is below 0.05. The exact way we compute the partial gamma coefficient for the two types of tests is described in Algorithm 1 in Appendix C. In the algorithm, A and B are arbitrary variables, but naturally they correspond to distinct items; if we check for LD and if we check for DIF, one is an item and one is an exogenous variable. All rejected tests are flagged and we move onto the next step.

2.1.4 Step 3

In this step of the screening process we begin the filtering of spurious evidence for LD and DIF.

Step 3(a) is a stepwise inclusion algorithm, which considers all the item pairs flagged for LD. For each item pair both directions are tested, and the described algorithm in Kreiner and Christensen (2011) argues that we should take the strongest evidence of LD at face value, and then disregard all other tests as spurious if one of the two items in the accepted genuine LD is defining the conditional rest-score. But how do we do this in practice? What if an LD test $Y_i \perp Y_{i'} \mid R_i$ is significant but the reverse direction $Y_{i'} \perp Y_i \mid R_{i'}$ is not? And how do we determine which evidence for LD is the strongest?

Step 3(a) in practice: From the initial screening of LD in step 2, we look at all flagged evidence of LD from which we may create four meaningful lists:

- A list containing all item pairs with two significant positive partial gamma coefficients.
- A list containing all item pairs with two significant negative partial gamma coefficients.
- A list with item pairs that only exhibit one positive partial gamma coefficient.

- A list with the item pairs that only exhibit one negative partial gamma coefficient.

For each item pair we need some summary statistic that we can use to evaluate the tests fairly across. Naturally, we compute the weighted partial gamma between the two, and use Algorithm 2 in Appendix C to determine which is genuine.

In this stepwise inclusion algorithm, we prioritize LD pairs which are significant in both directions before considering LD pairs with only one significant direction. Furthermore, we also allow for item pairs to drop from the set of two significant directions into the sets with only one, if a pair has been accepted as exhibiting genuine LD includes an item which is also used to define the rest-score for one of the tests in another item pair having two significant directions.

Example 2.1. Suppose we have a dataset with three items and the following lists of significant LD

Table 2: Item pairs with two significant positive partial γ coefficients.

item _i	item _{i'}	$\gamma_{ii'}^{(i)}$	$\gamma_{ii'}^{(i')}$	$WPG_{ii'}$
item 1	item 2	0.8	0.4	0.6
item 1	item 3	0.5	0.6	0.55

and

Table 3: Item pairs with only one significant positive partial γ coefficients.

item _i	item _j	$\gamma_{ii'}^{(i)}$	$\gamma_{ii'}^{(i')}$	$WPG_{ii'}$
item 2	item 3	0.8	0.1	0.45

Then, in the first iteration we would accept the evidence that local dependence between item 1 and item 2 is genuine. And the evidence that item 1 $\perp$ item 3 $| R_1$ with $\gamma = 0.5$ would be disregarded as spurious, hence the item pair 1 and 3 would be dropped to the set with only one significant positive partial correlation, and the LD between item 2 and 3 would be completely disregarded as its only significant direction is when conditioned on R_2 . So the two sets in the second iteration would be

Table 4: Item pairs with two significant positive partial γ coefficients.

item _i	item _{i'}	$\gamma_{ii'}^{(i)}$	$\gamma_{ii'}^{(i')}$	$WPG_{ii'}$
NA	NA	NA	NA	NA

and

Table 5: Item pairs with only one significant positive partial γ coefficients.

item _{<i>i</i>}	item _{<i>i'</i>}	$\gamma_{ii'}^{(i)}$	$\gamma_{ii'}^{(i')}$	WPG_{ij}
item 1	item 3	0.5	0.6	0.55

Thus, LD between item 1 and item 3 would be accepted as genuine, resulting in the two lists now being empty, and thus terminating the algorithm. The final set of what we believe to be genuine LD is {(item 1, item 2), (item 1, item 3)}.

◦

Step 3(b) is an elimination algorithm. Here we attempt to distinguish genuine sources of DIF, relative to items exhibiting it, from spurious. This step is implemented exactly as the procedure reads in Kreiner and Christensen (2011), which boils down to Algorithm 3 in Appendix C. Note, importantly, that no source is removed from the source list until all tests have been performed in the current iteration. Furthermore, in line 8 of that algorithm we refer to Algorithm 4 (also in Appendix C), which is in line with the discussion in Section 5.1 by Kreiner and Christensen (2011) on how to determine if the tests are significant. Algorithm 4 is simply a Monte Carlo based method for estimating p -values.

Important note: The partial gamma coefficient is used as an effect size measure. The Monte Carlo p -value is not obtained by permuting the partial gamma coefficient directly, but from the conditional independence test statistic implemented in `coin`. However, note that the two are obviously still related as they both assess conditional association between the same variables given the same strata, but they are not the same statistic. `Coin` uses some linear statistic, T , which is suggested by Strasser and Weber (1999), which is of the form:

$$T = \text{vec} \left(\sum_{i=1}^n w_i g(X_i) h(Y_i, (Y_1, \dots, Y_n))^T \right) \in \mathbb{R}^{pq}, \quad (4)$$

g is a transformation function of X and is referred to as the regression function, and h is a transformation of Y known as the influence function.

Step 3(c) The elimination algorithm applied in Step 3(c) is very similar to that used in 3(b). In fact, it is the exact same procedure, but instead of eliminating sources of spurious DIF, we eliminate items exhibiting spurious DIF w.r.t. exogenous variables. In other words, we are working with the $DIF(X_i)$ lists instead of the $SOURCE(Y_i)$ lists. Hence, we refer the curious reader to Algorithm 3 in Appendix C for the full algorithm.

Combine Step 3(b) and 3(c). As per the article Kreiner and Christensen (2011), the order of Step 3(b) and 3(c) does not make any difference since we are not making any final decisions until both steps have been performed. When both steps have been performed, we are left with a list of item and exogenous variable pairs, the partial gamma coefficient from testing independence between the pair given score, and possibly other items, and a p -value. Furthermore, for the same item and exogenous variable pair, we will have a partial gamma coefficient from the test of conditional independence between the pair given the score and possibly other exogenous variables together with the corresponding p -value. And not until now will final conclusions, on whether or not the DIF is genuine or not, be made. DIF will only be accepted as genuine if both p -values are significant.

Example 2.2. Assume we have five items $\{A, B, C, D, E\}$ and two exogenous variables $\{X, Z\}$. Suppose the initial DIF screening in Step 2 flags the following item-covariate pairs:

Item	Covariate	Partial γ	p -value
A	X	0.42	0.004
A	Z	0.31	0.030
C	X	-0.36	0.018
E	Z	0.55	0.001

Table 6: Initial evidence of DIF from Step 2.

From this table we construct the source and DIF sets

$$SOURCE(A) = \{X, Z\}, \quad SOURCE(C) = \{X\}, \quad SOURCE(E) = \{Z\},$$

and

$$DIF(X) = \{A, C\}, \quad DIF(Z) = \{A, E\}.$$

Step 3(b) investigates each item separately. For each item Y_i , we test whether each candidate source remains associated with Y_i after conditioning on the

score and the other candidate sources for that item. Suppose Step 3(b) gives the following final result:

Item	Candidate source	Hypothesis	p -value	3(b) decision	Iteration
A	X	$A \perp X \mid S, Z$	0.06	Retain	1
A	Z	$A \perp Z \mid S, X$	0.210	Remove	1
A	X	$A \perp X \mid S$	0.004	Retain	2
C	X	$C \perp X \mid S$	0.018	Retain	1
E	Z	$E \perp Z \mid S$	0.001	Retain	1

Table 7: Hypothetical Step 3(b) elimination of spurious DIF sources. Note that “Retain” in an intermediate iteration means that it will not be removed during that pass.

Thus Step 3(b) returns

$$SOURCE^{(3b)}(A) = \{X\}, \quad SOURCE^{(3b)}(C) = \{X\}, \quad SOURCE^{(3b)}(E) = \{Z\}.$$

Step 3(c) investigates each exogenous variable separately. For each covariate X_j , we test whether each candidate DIF item remains associated with X_j after conditioning on the score and the other candidate DIF items for that covariate. Suppose Step 3(c) gives the following final result:

Covariate	Candidate item	Hypothesis	p -value	3(c) decision	Iteration
X	A	$A \perp X \mid S, C$	0.09	Retain	1
X	C	$C \perp X \mid S, A$	0.160	Remove	1
X	A	$A \perp X \mid S$	0.004	Retain	2
Z	E	$E \perp Z \mid S, A$	0.2	Retain	1
Z	A	$A \perp Z \mid S, E$	0.280	Remove	1
Z	E	$E \perp Z \mid S$	0.001	Retain	2

Table 8: Hypothetical Step 3(c) elimination of spurious DIF items. Note that “Retain” in an intermediate iteration means that it will not be removed during that pass.

Thus Step 3(c) returns

$$DIF^{(3c)}(X) = \{A\}, \quad DIF^{(3c)}(Z) = \{E\}.$$

Finally, we combine the results from Step 3(b) and Step 3(c). An item-covariate pair is retained as genuine DIF only if it survives both procedures.

Item	Covariate	Survives Step 3.b	Survives Step 3.c	Conclusion
A	X	Yes	Yes	Genuine DIF
A	Z	No	No	Spurious
C	X	Yes	No	Spurious
E	Z	Yes	Yes	Genuine DIF

Table 9: Combining Step 3(b) and Step 3(c).

Therefore, the final genuine DIF relations are

$$A \sim X \quad \text{and} \quad E \sim Z.$$

Equivalently, the updated source and DIF sets are

$$SOURCE(A) = \{X\}, \quad SOURCE(E) = \{Z\},$$

and

$$DIF(X) = \{A\}, \quad DIF(Z) = \{E\}.$$

◦

2.1.5 Step 4

Moving away from DIF and LD for a moment, we are now interested in the structural association between the total score and exogenous variables, which we will investigate via backwards elimination search, as per Kreiner and Christensen (2011) suggestion. The idea here is to determine which exogenous covariates to retain as structural predictors of the score, which we do by tests of conditional independence between the score and exogenous variables given the remaining set of exogenous variables still retained.

Step 4 in practice. The general structure of Step 4 is clear from posed screening procedure. The backwards elimination search for exogenous variables associated with the score is implemented as described in Algorithm 5 in Appendix C. When we determine significance of tests, we do so using the same framework as in Step 3(b) and 3(c), i.e., via the Monte Carlo approximation implemented by Algorithm 4.

Retained covariates are treated as evidence of criterion validity, as they exhibit a significant conditional association with the total score. Dropped covariates, i.e., covariates with non-significant conditional independence tests, are interpreted as not contributing significant additional information towards the score after controlling for

other covariates in the candidate set, and should therefore not be included in the final model, as opposed to those we conclude to fulfill the criterion validity. The final result of Step 4 is therefore not just a collection of marginal score and exogenous variable associations. Each retained covariate has survived Algorithm 5, and we therefore allow ourselves to conclude that its association with the total score is non-trivial, and the covariate should therefore be included in the initial definition of the GLLRM in later steps.

2.1.6 Step 5

We are now ready to begin the initial construction of the GLLRM by collecting all the information from Step 3 and 4 outputs, and form the graph corresponding to the model, together with its corresponding IRT and moralized marginal graph.

Step 5 in practice We do not perform any new statistical tests in this step, but simply collect the information passed on from the previous steps and report the structure of the corresponding GLLRM graph. The GLLRM graph is quite straight forward, as we simply collect the genuine LD, genuine DIF, and which exogenous variables satisfy criterion validity with the latent trait. We report which nodes are present and where nodes are connected, beyond the edges connecting items to the latent trait. From this, we can determine how the IRT graph should look as it includes the same edges and nodes, but with item measurement, score association, and DIF edges being directed, while LD edges are undirected. Finally, the moralized marginal graph is obtained by marrying parents, replacing the latent trait with a score node, and including score-to-item edges and edges between all items, all of which are undirected.

2.1.7 Step 6

We are now ready to perform the last round of tests before reporting an initial model. In Step 5, we report a tentative model; now we subject this model to additional conditional independence tests for the otherwise retained DIF and LD in order to investigate if these relations still hold when taking the structure determined in Step 5 into account. In other words, we now check if LD and DIF edges found in Step 5 were only supported due to the simpler conditioning sets.

We perform three types of tests to determine this. For DIF, the tests are of the form

$$H_0^{C5} : Y_i \perp X_j \mid S, \text{DIF}(X_j) \setminus \{Y_i\}, \text{SOURCE}(Y_i) \setminus \{X_j\}. \quad (5)$$

For each LD edge we perform two tests. For generic item pair (Y_a, Y_b) exhibiting, what Step 3 has determined to be genuine LD, we now retest this in the following

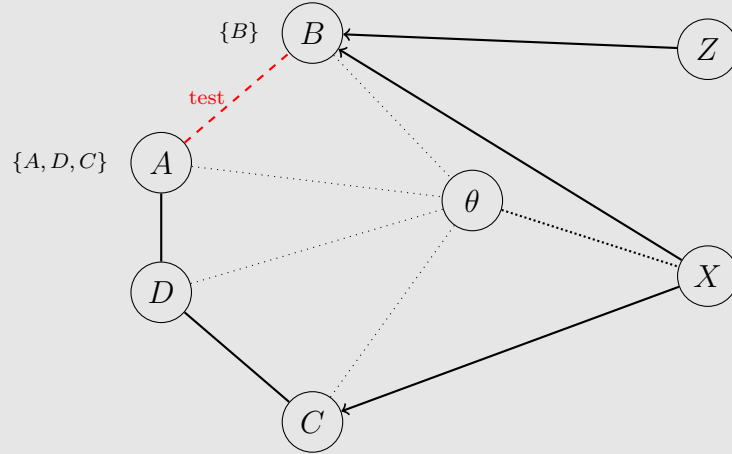
way: Separate the two pairs in the graph and observe which other LD connections are available from the two. We then consider the sub-score defined by the complement to the relevant sets, and use this to condition on together with other DIF sources for the corresponding LD connections. These are referred to as *C7* and *C9* hypothesis Kreiner and Christensen (2011). We allow the LD edge to be retained, if at least one of the two tests are significant.

Example 2.3. Assume we have a dataset with four items A, B, C, D and two exogenous covariates X, Z . Suppose within the dataset there exists DIF between the following pairs $(B, X), (C, X), (B, Z)$, LD between $(A, B), (C, D), (A, D)$, and finally X is related to the score S . Step 6 now retests the DIF using (5).

Item	Candidate source	Hypothesis	p -value	Decision
B	X	$B \perp X \mid S, C, Z$	0.016	Retain
C	X	$C \perp X \mid S, B$	0.210	Remove
B	Z	$B \perp Z \mid S, X$	0.04	Retain

Table 10: Hypothetical tests to determine if DIF edges should be retained.

Next, we check if the LD edges should be retained via the following procedure. Take the pair (A, B) , cut the edge between the two, and consider the LD connections still available from either one.



Temporarily testing LD edge (A, B) .

We clearly see from the graph above, that the LD connections still available

are

$$C_A = \{A, D, C\}, \quad C_B = \{B\}. \quad (6)$$

The tests we now perform are of conditional independence between the two items given the sub-scores defined by the complement set of items making up C_A and C_B , grouped with DIF sources relative to items defining the sub-scores.

$$C7: A \perp B \mid S_{C_A^c}, DIF(C_A) = A \perp B \mid S_{C_A^c}, X,$$

and

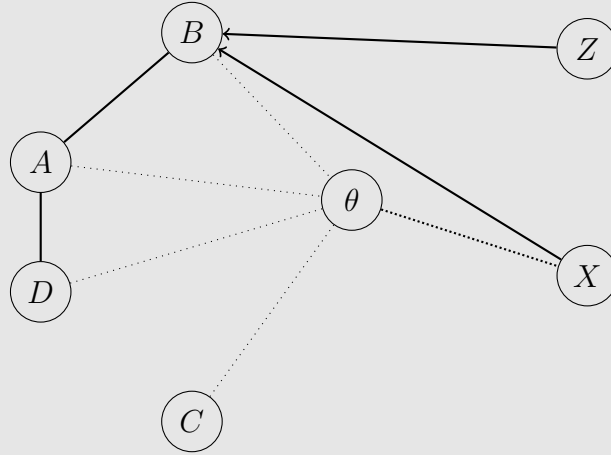
$$C9: A \perp B \mid S_{C_B^c}, DIF(C_B) = A \perp B \mid S_{C_B^c}, Z, X.$$

Doing this for all LD edges, suppose we observe the following table of tests

LD pair	Hypothesis	p -value	Decision
(A, B)	$B \perp A \mid S_{C_A^c}, X$	0.016	supported
(A, B)	$B \perp A \mid S_{C_B^c}, X, Z$	0.510	not supported
(C, D)	$C \perp D \mid S_{C_C^c}, X$	0.02	supported
(C, D)	$C \perp D \mid S_{C_D^c}, X, Z$	0.03	supported
(A, D)	$A \perp D \mid S_{C_A^c}, X, Z$	0.2	not supported
(A, D)	$A \perp D \mid S_{C_D^c}, X$	0.33	not supported

Table 11: Hypothetical tests to determine if LD edges should be retained.

Since an LD edge is removed if and only if the pair fails to achieve a significant p -value for both Step 6 LD tests, we conclude the two LD edges (A, B) and (A, D) are kept, while the edge between (C, D) is removed. Thus, we are left with the final Step 6 graph below.



Final Step 6 graph.

We illustrate the complete item screening procedure on a real dataset, the Shoulder Pain and Disability Index (SPADI), in Appendix D.

3 Discussion

The M2 and M3 checks are implemented as screening procedures measuring the associations between items and their corresponding rest-scores, and exogenous covariates and the total score respectively. This means we are not performing the exhaustive test of checking items and exogenous covariates against all possible sub-scores. This keeps the procedure computational feasible, as the number of unique sub-scores, for k items, not including the item in question and the 0 sub-score is 2^{k-1} , which grows exponentially with the number of items.

The stepwise inclusion algorithm, identifying genuine LD in step 3a, groups pairs with two significant LD tests and weigh them equally but prioritize them over item pairs only exhibiting one significant LD test, which are also grouped. In both groups the strongest evidence of genuine LD is determined by the absolute value of the weighted partial gamma coefficient of the two tests. This means that both positive and negative association LD can be prioritized as the strongest signal. The idea is that an item pair with two significant tests must have a stronger LD signal than item pairs with only one significant test. While allowing item pairs to drop from the set of two significant tests into the set with only one. The use of weighted partial gammas allow us to compare the LD pairs equally and also ensures the contribution of a test to the decision statistic is influenced proportionally to the number of comparable pairs defining the partial gamma, thus keeping as much information in the decision statistic as possible. Opposed to comparing pairs based on their naive arithmetic mean.

A negative partial gamma association between two items means that scoring high on one of the two will lead you to scoring low on the other. When does this makes sense? In a setting where two questions test different methods. But then we are no longer modeling a unidimensional latent trait, but a multi dimensional one Kreiner and Christensen (2004), and it should be well considered if the questionnaire should be handled as a whole or split into sub questionnaires.

For DIF, one of the two important decisions made, is that 3b and 3c are complementary steps and not individual, meaning no decisions are made until both have been performed and compared. We only retain a DIF edge if it survives both steps, which is a rather conservative choice. However, the general overall procedure is to start with a large set of LD and DIF (step 2) and then proceed to narrow the sets down as we get deeper into the screening. As all LD and DIF edges are tested again

in later steps. The advantage being that we are less likely to include spurious DIF edges, but at the cost of potentially removing weaker but genuine DIF relations. The second important of the two decisions made in the DIF framework is how we estimate p -values. As strata quickly become so thin that asymptotic p -values are unreliable, we've chosen to rely on Monte Carlo estimation of the exact p -values. While Monte Carlo methods may provide an easy framework for approximating p -values it also introduces some simulation uncertainty. In other words, the number of resamples B affects the stability and therefore precision of p -values, not to mention how the unique resampling actually looks for a specific p -value. Alternative methods include; asymptotic p -values (which should be noted: is implemented for reproducibility purposes!), bootstrapping and/or other resampling procedures. Asymptotic tests are faster, almost instantaneous, but may be less reliable in large sparse contingency tables, while bootstrapping or resampling methods may be more computationally demanding tasks and require additional choices to be made about what exactly we want to resample.

Finally, the graph-building and graph-checking steps, i.e. 5 and 6, does not draw the graphs explicitly, this is something we have, as of now, left to the user. They do however report the nodes, where edges should be present, which type they are and if they should be directed. Step 5 reports this information based on the retained LD, DIF, and score-covariate evidence. It should be stressed, the graph structure reported, is to be understood as an evidence graph which is the result of the previous steps and not as the final model. Step 6 then retests the retained edges using conditioning sets implied by the graph. Meaning that right now step 6 acts as a confirmation of the posed graph from step 5.

4 Conclusion

This project has shown that the item screening procedure within the framework of graphical log-linear Rasch models can be implemented in R. The result is an R package that provides a coherent and reproducible toolkit for carrying out such screenings. The package addresses the need for a more transparent workflow for researchers working with Rasch models and its corresponding assumptions.

4.1 Beyond twigg

A natural extension of twigg would be to include formal tests of minimal GMP hypotheses. This would make it possible for the package to support not only the screening procedure itself, but also later stages of model evaluation. However, this is not essential for the current contribution of the package. In its present form, twigg should be understood as a tool for carrying out and documenting the screening process, while

the final interpretation and subsequent analysis remain the responsibility of the user.

Future development could also focus on improving the integration between twigg and existing Rasch model packages, adding clearer reporting tools, and providing more extensive graphical summaries of the screening results. Such extensions would further support transparency, reproducibility, and practical usability for researchers applying graphical log-linear Rasch models.

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Appendix A

Measure of Association

The Goodman and Kruskal (1954) γ measures the association between two ordinal variables. After cross-tabulation it looks at pairs of observations, counting the total number of concordant pairs (pairs that are ranked the same way by the two variables) C , and the total number of discordant pairs (pairs that are ranked differently by the two variables) D . The value is

$$\gamma^{GK} = \frac{C - D}{C + D}. \quad (7)$$

The partial γ coefficient Davis (1967) is computed this partial gamma, we will need to determine which variable we are controlling for, and hence conditioning on. Let's say we are testing for DIF between Y_i and X_j , then we need to condition on the total score. So, the partial gamma coefficient is computed by first stratifying the data with respect to each unique value of the score. Let n denote the total number of strata, then for each strata we compute Goodman and Kruskal's gamma, while also storing the number of concordant and discordant pairs that define each GK gamma, and we can compute the partial gamma coefficient as a weighted sum of the strata-specific GK gamma's.

$$\hat{\gamma}_{Y_i \perp X_j | S} = \sum_{i=1}^n w_i \gamma_i^{GK} = \sum_{i=1}^n \frac{(C_i + D_i)}{\sum_{i=1}^n (C_i + D_i)} \frac{C_i - D_i}{C_i + D_i} \quad (8)$$

$$= \frac{\sum_{i=1}^n (C_i - D_i)}{\sum_{i=1}^n (C_i + D_i)} = \frac{\sum_{i=1}^n C_i - \sum_{i=1}^n D_i}{\sum_{i=1}^n C_i + \sum_{i=1}^n D_i}. \quad (9)$$

The weights, and hence the contribution per GK gamma, is the ratio between the sum of concordant and discordant pairs that defines the i th γ^{GK} , and the sum of all strata-specific concordant and discordant pairs. This allows the individual γ^{GK} to contribute a weight towards the partial gamma that is proportional to the number of observations defining that γ^{GK} , Agresti (1984).

The core logic is the same behind both statistics, since we collect the number of concordant and discordant pairs and calculate the ratio between the difference and the sum. Hence, both entities lie in the closed set $[-1, 1]$, and naturally the closer we observe the association measure to one of the two end-points the stronger the relationship Agresti (1984).

Appendix B

B.1 The Multiple Testing Problem

When we perform the second step of the screening procedure (see table 1 in section 2) we are evaluating a lot of tests at once, which means the risk of false discovery rate (FDR) increases. This will need to be accounted for.

B.1.1 The Benjamin-Hochberg Procedure

A method for controlling the FDR, when performing multiple hypothesis tests at once, is via the Benjamin-Hochberg procedure. This procedure adjusts the raw p -values to account for multiple testing, resulting in a more conservative set of accepted alternative hypotheses. The BH procedure is quite simple. Say we perform T tests, the T raw p -values are ranked in ascending order $p_1 < p_2 < \dots < p_T$. The BH p -adjusting method then proceeds to control the FDR at the level we determine the raw p -values significance level, α , by finding the largest t such that $p_t \leq \frac{t}{T}\alpha$ and label all p_i significant for $i = 1, \dots, t$, Benjamini and Hochberg (1995).

Appendix C

Algorithms

This appendix collects the pseudocode for the algorithms implementing the item screening procedure described in Section 2, in the order in which they are used.

Step 2: Partial Gamma Coefficient

Algorithm 1 Partial γ coefficient for testing $A \perp B \mid \mathcal{C}$

Require: Dataset with item responses and exogenous covariates; variables A and B ; conditioning variables $\mathcal{C}$

Ensure: Partial gamma coefficient $\gamma_{A \perp B \mid \mathcal{C}}$

Initialize

$$C \leftarrow 0, \quad D \leftarrow 0.$$

Stratify the dataset according to the joint levels of the conditioning variable(s) $\mathcal{C}$.

Let n denote the number of resulting strata.

for $l = 1, \dots, n$ **do**

 Let $\mathcal{D}_l$ denote the observations in stratum l .

if $\mathcal{D}_l$ contains fewer than two observations **then**

Skip stratum l .

Construct the contingency table of A and B within stratum l .

if the contingency table has fewer than two rows or fewer than two columns
then

Skip stratum l .

Compute the number of concordant pairs in stratum l :

$$C_l \leftarrow \#\{\text{concordant pairs in stratum } l\}.$$

Compute the number of discordant pairs in stratum l :

$$D_l \leftarrow \#\{\text{discordant pairs in stratum } l\}.$$

Update

$$C \leftarrow C + C_l, \quad D \leftarrow D + D_l.$$

if $C + D = 0$ **then**
return NA

Compute

$$\gamma_{A \perp B | \mathcal{C}} \leftarrow \frac{C - D}{C + D}.$$

return $\gamma_{A \perp B | \mathcal{C}}$

Step 3(a): Elimination of Spurious Local Dependence

Algorithm 2 Elimination of Spurious Local Dependence

Require: Set of significant local dependence hypotheses from the initial LD screening. For each item pair $\{Y_i, Y_{i'}\}$, at most two directional hypotheses can be significant:

$$H_{ii'}^{(i)} : Y_i \perp Y_{i'} \mid R_i, \quad H_{ii'}^{(i')} : Y_i \perp Y_{i'} \mid R_{i'}.$$

Let $\gamma_{ii'}^{(i)}$ and $\gamma_{ii'}^{(i')}$ denote the corresponding partial γ coefficients, and let $w_{ii'}^{(i)}$ and $w_{ii'}^{(i')}$ denote the corresponding numbers of comparable pairs.

Ensure: Reduced set $\mathcal{G}$ of item pairs considered to exhibit genuine local dependence.

1: Initialize the set of active significant directional LD hypotheses:

$$\mathcal{H} \leftarrow \left\{ H_{ii'}^{(k)} : H_{ii'}^{(k)} \text{ is significant, } k \in \{i, i'\} \right\}.$$

2: Initialize the set of genuine LD relations:

$$\mathcal{G} \leftarrow \emptyset.$$

- 3: For each item pair $\{Y_i, Y_{i'}\}$ represented in the initial LD screening, compute the pair-level weighted partial gamma

$$WPG_{ii'} = \frac{w_{ii'}^{(i)} \gamma_{ii'}^{(i)} + w_{ii'}^{(i')} \gamma_{ii'}^{(i')}}{w_{ii'}^{(i)} + w_{ii'}^{(i')}}.$$

- 4: **while** $\mathcal{H} \neq \emptyset$ **do**

- 5: Identify all unordered item pairs $\{Y_i, Y_{i'}\}$ represented by at least one hypothesis in the current active set $\mathcal{H}$.
- 6: Using only the currently active significant directional hypothesis, partition the represented item pairs into four evidence lists:

$$\mathcal{L}_{++}^{(2)} = \left\{ \{Y_i, Y_{i'}\} : H_{ii'}^{(i)} \in \mathcal{H}, H_{ii'}^{(i')} \in \mathcal{H}, \gamma_{ii'}^{(i)} > 0, \gamma_{ii'}^{(i')} > 0 \right\},$$

$$\mathcal{L}_{--}^{(2)} = \left\{ \{Y_i, Y_{i'}\} : H_{ii'}^{(i)} \in \mathcal{H}, H_{ii'}^{(i')} \in \mathcal{H}, \gamma_{ii'}^{(i)} < 0, \gamma_{ii'}^{(i')} < 0 \right\},$$

$$\mathcal{L}_+^{(1)} = \left\{ \{Y_i, Y_{i'}\} : \text{exactly one of } H_{ii'}^{(i)}, H_{ii'}^{(i')} \text{ is in } \mathcal{H} \text{ and its partial } \gamma \text{ is positive} \right\},$$

$$\mathcal{L}_-^{(1)} = \left\{ \{Y_i, Y_{i'}\} : \text{exactly one of } H_{ii'}^{(i)}, H_{ii'}^{(i')} \text{ is in } \mathcal{H} \text{ and its partial } \gamma \text{ is negative} \right\}.$$

- 7: First, consider pairs with two active significant partial correlations:

$$\mathcal{L}^{(2)} = \mathcal{L}_{++}^{(2)} \cup \mathcal{L}_{--}^{(2)}.$$

- 8: **if** $\mathcal{L}^{(2)} \neq \emptyset$ **then**

- 9: Select the pair with the largest absolute weighted partial gamma among pairs with two active significant partial correlations:

$$\{Y_a, Y_b\} = \arg \max_{\{Y_i, Y_{i'}\} \in \mathcal{L}^{(2)}} |WPG_{ii'}|.$$

- 10: **else**

- 11: Select the pair with the largest absolute weighted partial gamma among pairs with only one active significant partial correlation:

$$\{Y_a, Y_b\} = \arg \max_{\{Y_i, Y_{i'}\} \in \mathcal{L}_+^{(1)} \cup \mathcal{L}_-^{(1)}} |WPG_{ii'}|.$$

- 12: Add the selected item pair to the set of genuine LD relations:

$$\mathcal{G} \leftarrow \mathcal{G} \cup \{\{Y_a, Y_b\}\}.$$

- 13: Remove from the active hypothesis set, all directional LD hypotheses whose rest-score is defined by either of the selected items:

$$\mathcal{H} \leftarrow \mathcal{H} \setminus \left\{ H_{ii'}^{(k)} \in \mathcal{H} : k \in \{a, b\} \right\}.$$

- 14: Thus, if an item pair previously had two active significant directional hypotheses, but only one of them remains after this removal, the pair may be reconsidered in a later iteration in either $\mathcal{L}_+^{(1)}$ or $\mathcal{L}_-^{(1)}$.

- 15: **return** $\mathcal{G}$
-

Step 3(b): Elimination of Spurious DIF Sources

Algorithm 3 Elimination of Spurious DIF Sources for Item Y_i

Require: Dataset with item responses and exogenous covariates; item Y_i ; item set $\mathcal{I}$; initial source set $\text{SOURCE}(Y_i)$; and significance level α

Ensure: Final set of genuine DIF sources $\widehat{\text{SOURCE}}(Y_i)$.

- 1: Compute the total score

$$S = \sum_{Y_k \in \mathcal{I}} Y_k.$$

- 2: Initialize

$$\mathcal{S}_i^{(0)} \leftarrow \text{SOURCE}(Y_i).$$

- 3: Set $t \leftarrow 0$.

- 4: **repeat**

- 5: Let

$$\mathcal{P}_i^{(t)} \leftarrow \mathcal{S}_i^{(t)},$$

denote the source set used during the current pass.

- 6: Set

$$X_{\text{drop}} \leftarrow \emptyset, \quad p_{\text{drop}} \leftarrow -\infty.$$

- 7: **for all** $X_j \in \mathcal{P}_i^{(t)}$ **do**

- 8: Use Algorithm 4 to test

$$H_0 : Y_i \perp X_j \mid S, \mathcal{P}_i^{(t)} \setminus \{X_j\}.$$

- 9: Let $p_j^{(t)}$ denote the corresponding p -value.

- 10: **if** $p_j^{(t)} > \alpha$ and $p_j^{(t)} > p_{\text{drop}}$ **then**

- 11: Set

$$X_{\text{drop}} \leftarrow X_j, \quad p_{\text{drop}} \leftarrow p_j^{(t)}.$$

```

12:   if  $X_{\text{drop}} = \emptyset$  then
13:       Stop the elimination procedure.
14:   else
15:       Remove the non-significant source with the highest  $p$ -value found in the
       current pass:

$$\mathcal{S}_i^{(t+1)} \leftarrow \mathcal{S}_i^{(t)} \setminus \{X_{\text{drop}}\}.$$

16:       Set  $t \leftarrow t + 1$ .
17:   until no source is removed or  $\mathcal{S}_i^{(t)} = \emptyset$ 
18:   Set

$$\widehat{\text{SOURCE}}(Y_i) \leftarrow \mathcal{S}_i^{(t)}.$$

19: return  $\widehat{\text{SOURCE}}(Y_i)$ 

```

Monte Carlo Approximation of Conditional Independence Tests

Algorithms 3 and 5 both rely on the following Monte Carlo test of conditional independence, which is also used directly in Step 3(c).

Algorithm 4 Monte Carlo approximation of the p -value for a conditional independence test

Require: Dataset with item responses and exogenous covariates; variables Y_i and X_j ; conditioning variables $\mathcal{C}$; number of Monte Carlo samples B

Ensure: Observed partial γ coefficient and Monte Carlo p -value for testing

$$H_0 : Y_i \perp X_j \mid \mathcal{C}.$$

- 1: Restrict the dataset to complete cases on $\{Y_i, X_j\} \cup \mathcal{C}$.
- 2: Use Algorithm 1 to compute the observed partial gamma coefficient

$$\widehat{\gamma}_{obs} = \gamma_{Y_i \perp X_j \mid \mathcal{C}}.$$

- 3: Create a stratification variable by taking the joint levels of the conditioning variables $\mathcal{C}$.
- 4: Remove all strata with fewer than two observations.
- 5: Remove all strata in which either Y_i or X_j has no variation.
- 6: Recompute the stratum sizes and again remove strata with fewer than two observations.
- 7: **if** no usable strata remain **then**
- 8: **return** $(\widehat{\gamma}_{obs}, \text{NA})$
- 9: Define the conditional independence test

$$H_0 : Y_i \perp X_j \mid \text{strata}.$$

- 10: Using the conditional independence framework implemented in `coin`, construct the test statistic for association between Y_i and X_j conditional on the strata.
- 11: Approximate the conditional null distribution of the test statistic by drawing B Monte Carlo samples from the null distribution, where permutations are performed conditionally within strata.
- 12: Let T_{obs} denote the observed test statistic, and let

$$T^{(1)}, \dots, T^{(B)},$$

denote the Monte Carlo samples from the conditional null distribution.

- 13: Compute the two-sided Monte Carlo p -value as the proportion of sampled null statistics at least as extreme as the observed statistic:

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}(|T^{(b)}| \geq |T_{obs}|).$$

- 14: **return** $(\hat{\gamma}_{obs}, p)$
-

Step 4: Structural Screening of Covariates

Algorithm 5 Step 4 structural screening

Require: Dataset with item responses and candidate covariates; significance level α ; number of Monte Carlo samples B

Ensure: Final set of covariates which satisfy criterion validity.

- 1: Compute the total score S as the row sum of the item responses.
- 2: Let $\mathcal{C}$ be the set of candidate covariates.
- 3: Let $\mathcal{R} \leftarrow \emptyset$ be the set of removed covariates.
- 4: **while** $\mathcal{C}$ is not empty **do**
- 5: **for** each covariate $X_j \in \mathcal{C}$ **do**
- 6: Test

$$H_0 : S \perp X_j \mid \mathcal{C} \setminus \{X_j\},$$

using a conditional partial gamma test

- 7: Store the partial gamma estimate and the corresponding p -value.
- 8: **if** all decision p -values are less than or equal to α **then**
- 9: Stop and retain all covariates in $\mathcal{C}$.
- 10: **else**
- 11: Select the covariate X_j with the highest decision $p_j > \alpha$.
- 12: Remove X_j from $\mathcal{C}$.
- 13: Add X_j to $\mathcal{R}$.

- 14: **return** retained covariates $\mathcal{C}$ and removed covariates $\mathcal{R}$
-

Appendix D

The SPADI Questionnaire: A Worked Example

This appendix applies the full item screening procedure of Section 2 to a real dataset: the Shoulder Pain and Disability Index (SPADI).

D.1 Step 0: Getting Started

The screening procedure does not contain a Step 0 explicitly, but naturally it is expected that the statistician gets familiar with the data at hand and potentially pre-processes it before any screening and/or analysis is performed.

The dataset contains observations from participants answering a questionnaire with 13 questions in total, five concerning pain and eight concerning disabilities. All items are scored on the discrete scale, $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$, with a higher number indicating a higher level of pain or disability associated with the given item. Beyond the item scores two exogenous variables “gender” and “above60” are present. The total number of observations available is 229. From the 229 observations, 213 are complete with respect to item responses. The empirical distribution of missing values for the two exogenous variables, in the subset of complete item responses, is 0 for both. Hence, we will be working with the entire 213 cases throughout the screening. As per Christensen et al. (2019), we will be splitting the dataset and treating the two sub-scales, pain and disability, as two different questionnaires, while also adhering to the proposal that we should map the possible scores from $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$ to $(0, 0, 1, 1, 2, 2, 3, 3, 4, 4, 5)$. We will therefore move forward with two datasets, a 213×7 pain questionnaire and a 213×10 disability questionnaire.

D.2 Step 1

Both sub-scales pass the M1 and M2 checks, the results are not shown here, but can be seen in Appendix B. The exogenous variable “over60” fail the M3 check for both sub-scales (see Appendix B). This means we should re-consider if it makes sense to continue. It would not, if we consider “over60” a criterion variable, however we do not. Thus we proceed.

D.3 Step 2

During this step, we perform multiple tests. Naturally, we will control for the FDR using Benjamin-Hochbergs procedure. The disability sub-scale does not show any sign of DIF during this initial step of screening, however the pain sub-scale does. Item *P1*

and the exogenous covariate “over60” exhibit DIF with ($\gamma = -0.43$, $p < 0.001$). For the LD screening, the two sub-scales does exhibit some. The disability sub-scale has one instance of LD, between the pair ($D4, D5$) for which both directions are significant. The pain sub-scale has two pairs exhibiting LD; ($P3, P4$), which is significant in both directions, and ($P1, P4$), which is only significant when conditioning on R_{P4} .

D.4 Step 3

D.4.1 Step 3(a)

After the inclusion algorithm, we find the following pairs of genuine LD in the two sub-scales

Disability sub-scale		
Item1	Item2	partial γ
$D4$	$D5$	0.425

Table 12: Genuine LD pairs.

Pain sub-scale		
Item1	Item2	partial γ
$P3$	$P4$	0.467

Table 13: Genuine LD pairs.

D.4.2 Step 3(b) and 3(c)

As the initial screening of DIF for the disability sub-scale did not disclose any evidence of DIF, we will only focus on the pain sub-scale in these two steps. Furthermore, we will be using the asymptotic p -values of the partial gamma coefficients for reproducibility purposes, and to stay as transparent as possible. However, since we only found one instance of DIF in the pain sub-scale, the elimination algorithm will report this exact DIF relation and terminate.

item	source	partial γ_1	$\mathcal{C}_1$	p_1	partial γ_2	$\mathcal{C}_2$	p_2	conclusion
$P1$	over60	-0.430	Score	0.0009	-0.430	Score	0.0009	DIF

Table 14: Test for genuine DIF. With $\mathcal{C}_i$ denoting the conditioning set of variables for the relevant partial γ_i .

D.5 Step 4

For the disability sub-scale, both exogenous covariates support criterion validity as none of the two are independent of the score given the other. For the pain sub-scale, it’s a different story, none of the two support criterion validity.

Subscale	covariate	partial γ	p	$\mathcal{C}$	S.P.C.
Disability	gender	0.248	0.0007	over60	TRUE
	over60	0.190	0.027	gender	TRUE
Pain	gender	0.080	0.115	over60	FALSE
	over60	0.029	0.860	gender	FALSE

Table 15: Again $\mathcal{C}$ is the conditioning set and S.P.C. is an abbreviation for “supports criterion validity”.

D.6 Step 5

We are now ready to construct the graph for the two sub-scales.

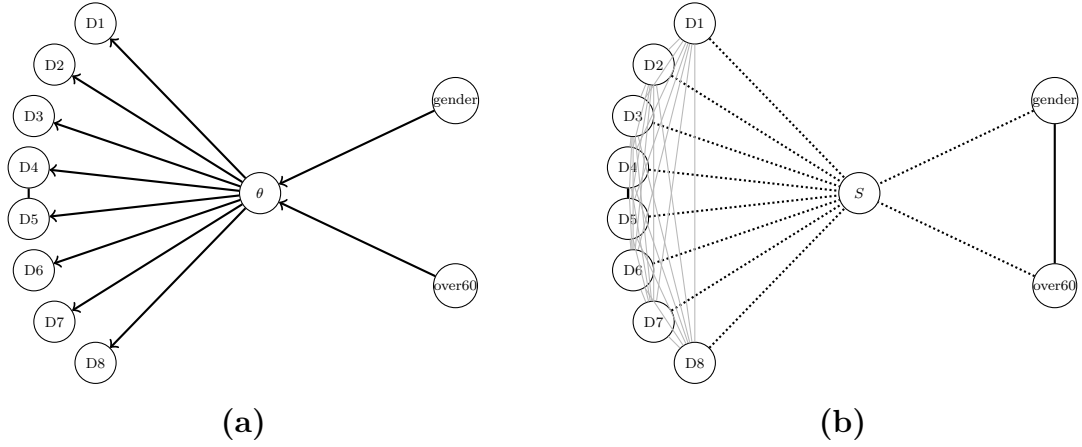


Figure 1: Disability sub-scale graphs. (a) Shows the IRT graph and (b) the corresponding moralized graph.

D.7 Step 6

In this step we want to retest the DIF and LD edges but conditioned on greater sets. However, due to the structure of LD and DIF in the two sub-scales, these tests are redundant and we may conclude the final graphs to be the same as those reported in Step 5, see Figures 1 and 2.

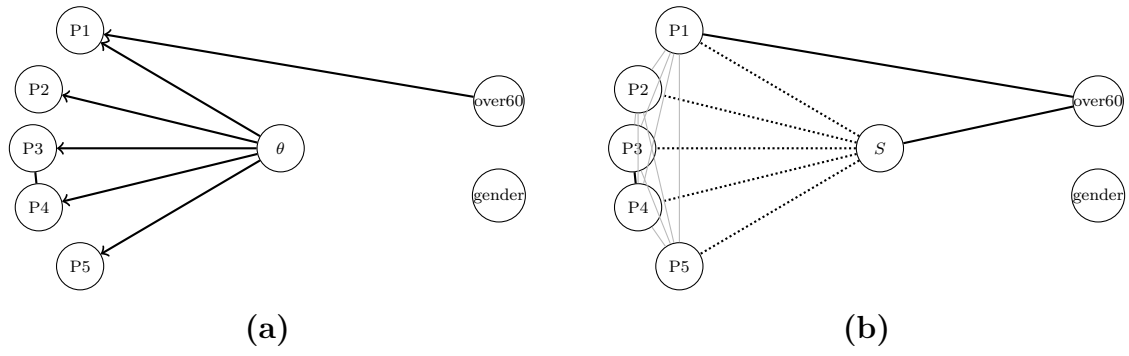


Figure 2: Pain sub-scale graphs. (a) Shows the IRT graph and (b) the corresponding moralized graph.